

Your birthday: **dd/mm**

Let **d** = sum of digits of **dd** and **m** = sum of digits of **mm**.

Question 1. (10 points) The weights of certain machine components are normally distributed with a mean of $(80 + \mathbf{m})$ g and a standard deviation of **d** g. Find the two weights that separate the top **d**% and the bottom **m**%. (These weights could serve as limits used to identify which components should be rejected.)

Question 2. (10 points) A study of the amount of time it takes a mechanic to rebuild the transmission for a new model of Chevrolet Cavalier shows that the mean is $(8 + 1/\mathbf{d})$ hours and the standard deviation is $(2 - 1/\mathbf{m})$ hours. If $40 + \mathbf{m}$ mechanics are randomly selected, find the probability that their mean rebuild time is less than $(8 - 1/\mathbf{d})$ hours.

Question 3. (10 points) In a survey of 5400 T.V. viewers, 30% said they watch network news programs. Find the **margin of error** for the 95% confidence interval used to estimate the population proportion.

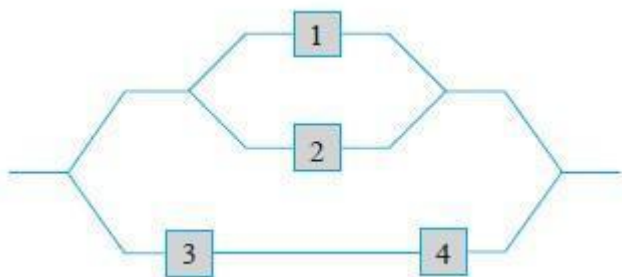
Question 4. (10 points) A laboratory tested $80 + \mathbf{m}$ chicken eggs and found that the mean amount of cholesterol was $200 + \mathbf{m}$ milligrams with the standard deviation $s = (10 + \mathbf{d}/10)$ milligrams. Construct a $(90 + \mathbf{m})$ percent confidence interval for the true mean cholesterol content, μ , of all such eggs.

Question 5. (10 points) Consider the system of components connected as in the accompanying picture. Components 1 and 2 are connected in parallel, so that subsystem works iff either 1 or 2 works; since 3 and 4 are connected in series, that subsystem works iff both 3 and 4 work. If components work independently of one another and

$$P(\text{component 1 works}) = P(\text{component 2 works}) = 1 - \mathbf{d}/40$$

$$P(\text{component 3 works}) = P(\text{component 4 works}) = 1 - \mathbf{m}/20$$

Calculate $P(\text{system works})$.



Question 6. (10 points) A recent study indicated that $(20 + \mathbf{d})\%$ of the $100\mathbf{m}$ women over age 55 in the study were widows.

a. How large a sample must you take to be $(90 + \mathbf{m})\%$ confident that the estimate is within $\mathbf{m}/200$ of the true proportion of women over age 55 who are widows?

b. If no estimate of the sample proportion is available, how large should the sample be?

Using data: <https://docs.google.com/spreadsheets/d/1tfGZLMuc5XasBjfiTDcIZakoGuQ-oL1cPw4f8lhFIQY/edit?usp=sharing>

For 2 question 7 & 8.

Question 7. (20 points) Find the equation for this data in 2 models:

- a) Linear
- b) Quadratic

Choose the better model and predict the next 5 values.

Question 8. (20 points) Given the time-series (in quarter) in the data. Find the model and predict the next 4 values.